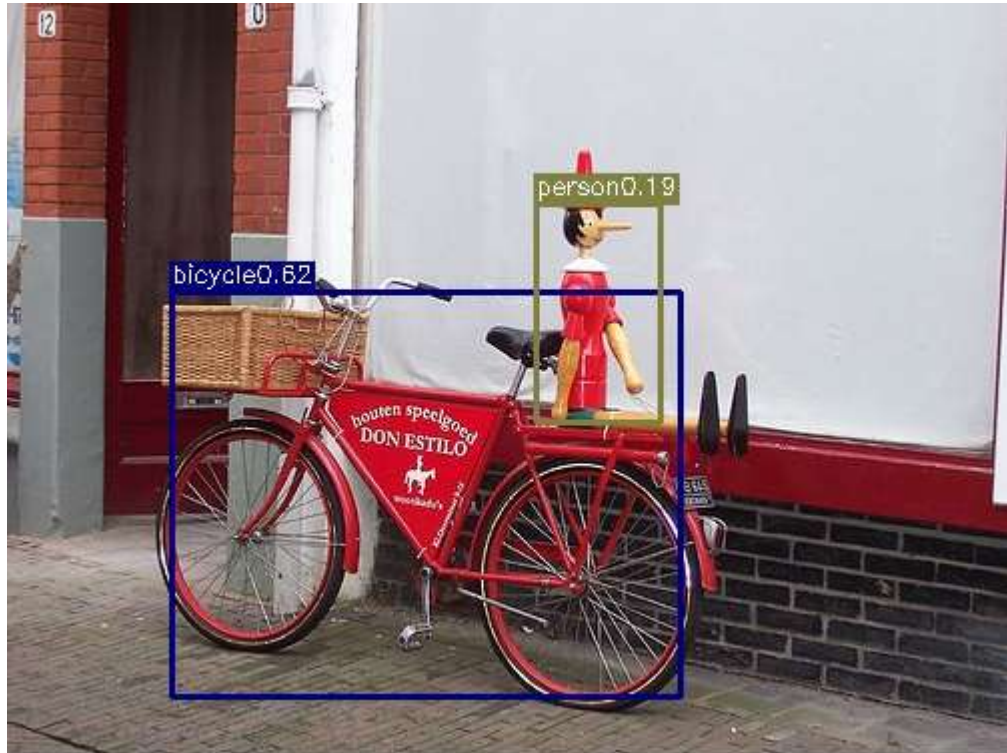


Bishop's University
CS 596 – Special Topics on Deep Learning
Final Exam: YOLO Object Detection on
PASCAL VOC



In this final exam you will implement a YOLO-like object detector on the [PASCAL VOC 2007 dataset](#) to produce results like in the above image. The goal is to help you understand the fundamentals of training an object detector, gain experience with PyTorch, and learn how to use pre-trained models provided by the deep learning community.

How to start

You have to rely on the provided starting code. The top-level notebook (*MP3_P2.ipynb*) will guide you through all of the steps. You will mainly focus on implementing the loss function of YOLO in the `yolo_loss.py` file. You will be provided a pre-trained network structure for the model. The network structure has been inspired by [DetNet](#), however you are not required to understand it. In principle, it can be replaced by a different network

architecture and trained from scratch, but to achieve a good accuracy with a minimum of computational expense and tuning, you should stick to the provided one.

As you start this part, you will realize that this is a more computationally intensive assignment than what you are used to. To get an idea whether your implementation works without waiting a long time for training to converge, here are some average values to expect:

Epoch Test loss

1	5.25
2	4.62
5	3.66
10	3.21

Epoch mAP

5	0.0000
10	0.2412
15	0.3754
20	0.4096
25	0.4823
30	0.4892
35	0.5092
40	0.5049
45	0.5109
50	0.5088

To train this model in a reasonable amount of time, you'll need to use a GPU. This can either be your personal GPU, Google Colab, or Google Cloud Platform.

Here are two interesting links about Google Colaboratory with the instructions to install Pytorch:

<https://towardsdatascience.com/getting-started-with-google-colab-f2fff97f594c>

<https://colab.research.google.com/notebooks/welcome.ipynb>

By this environment, you will be able to do GPU programming and to run your programs through the cloud computing provided by Google.

Environment Setup (Local)

If you will be completing the assignment on a local machine then you will need a Python environment set up with the appropriate packages.

We suggest that you use [Anaconda](#) to manage python package dependencies. This [guide](#) provides useful information on how to use Conda.

Data Setup (Local)

Once you have downloaded the zip file, create a folder Project and execute the download_data script provided::

```
/download_data.sh
```

Data Setup (For Colaboratory)

If you are using Google Colaboratory for this final exam you will need do some additional setup steps.

You will need to run the download_data.sh script inside colab cells and move all the data to a local directory in colab for fast access.

You will now need to open the final project ipython notebook file from your Google Drive folder in Colaboratory and run a few setup commands. Make sure to set the GPU as the hardware accelerator. To do this, on the top bar choose Edit->Notebook Settings->select 'GPU'. We have condensed all the important setup commands you need to run into an ipython notebook [here](#).

IPython

The final project is given to you in the file *MP3_P2.ipynb*. As mentioned, if you are using Colaboratory, you can open the ipython notebook directly in Colaboratory. If you are using a local machine, ensure that ipython is installed (<https://ipython.org/install.html>). You may then navigate the assignment directory in terminal and start a local ipython server using the jupyter notebook command.

Resources

- [Lecture on object detection](#),
- [Original YOLO paper](#),
- [Great post about YOLO on Medium](#),

- [Differences between YOLO, YOLOv2 and YOLOv3](#),
- [Great explanation of the Yolo Loss function](#).

Submission

You will have to submit all the programmed solutions. Please, provide a pdf report that highlights the team members, the contribution of each one of them, and the obtained results. All of your code (python files and ipynb file) in a single ZIP file. In addition, the filename should be **netid_mp3_code.zip**. **Do NOT include datasets in your zip file.** Your ipython notebook with output cells converted to PDF format. The filename should be **netid_mp3_output.pdf**. In your pdf report, you have to specify the following:

1. Contributions of you and your colleagues,
2. The final loss value:
3. What did not work in your code (if anything):
4. Sample Images (four images) from your detector from PASCAL VOC:
5. Explanation of the obtained results.

Good luck :)